
AWS IAM Identity Center (AWS SSO) — Explained

What is IAM Identity Center?

AWS IAM Identity Center (formerly AWS Single Sign-On) is a **centralized identity and access management service** that enables you to manage **user identities** and **permissions across AWS accounts** and **integrated applications** easily.

👉 It's **ideal for organizations** using **multiple AWS accounts** (via AWS Organizations) or using third-party apps like Salesforce, Microsoft 365, etc.

Key Features

Feature	Description
 Centralized Access	Manage access to multiple AWS accounts from a single place.
 Connect External IdPs	Integrate with Microsoft AD, Okta, Azure AD, etc.
 User Portal	Users get a personalized portal to access their assigned accounts/apps.
 Fine-Grained Permissions	Assign IAM roles using permission sets.
 Just-in-Time Provisioning	Automatically create users in AWS from your IdP.

Lab Setup: IAM Identity Center + AWS Organization

Pre-requisites:

- A **management account** of AWS Organization

- At least one **member AWS account**
 - AWS IAM Identity Center **enabled** in a Region (e.g., [us-east-1](#))
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Step-by-Step Lab

Step 1: Enable IAM Identity Center

1. Go to **AWS Console > IAM Identity Center**.
 2. Click **Enable**.
 3. Choose the **region** (e.g., [us-east-1](#)).
 4. It automatically integrates with **AWS Organizations** if you are using it.
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Step 2: Choose Identity Source

You can pick from:

- **AWS IAM Identity Center** (built-in)
- **External Identity Provider (IdP)** (e.g., Azure AD, Okta)
- **Active Directory** (AWS Managed AD or On-Premises)

 For now, choose: [AWS IAM Identity Center](#) (default).

Step 3: Create Users & Groups

1. Go to **Users** > Click **Add user**.
 2. Add a user like `student1@example.com`.
 3. Add a group like `Developers`, and assign users.
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Step 4: Create Permission Sets

1. Go to **Permission sets** > **Create permission set**.
 2. Choose a template like:
 - `PowerUserAccess`
 - `ReadOnlyAccess`
 - Or create **custom JSON policies**
 3. Name it like `DeveloperAccess`.
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Step 5: Assign Users to AWS Accounts

1. Go to **AWS accounts** tab.
 2. Select one or more accounts.
 3. Click **Assign users or groups**.
 4. Choose `student1` and assign `DeveloperAccess`.
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Step 6: Login to User Portal

- Visit the **User Portal URL** (found in the IAM Identity Center dashboard).
- Login as **student1**.
- You'll see access to AWS accounts and roles.

 Boom! You're in. The user can switch roles or access AWS accounts as assigned.

Integration with External Identity Providers (Bonus)

You can also link your **Azure AD** or **Okta** using **SAML 2.0**.

- Go to **Settings > Identity Source > External identity provider**
 - Upload metadata from your IdP
 - Configure attribute mappings (email, first name, etc.)
 - Done! 
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Summary

Concept	Description
 IAM Identity Center	Unified identity & access across AWS
 Identity Source	Where users come from (AWS, AD, or external)
 Permission Set	IAM role templates applied to users
 Role Assignment	Grant users/groups access to AWS accounts



User Portal

One-stop dashboard for users

Use Cases

- Centralized access management across multiple AWS accounts
 - Student portals for AWS education
 - Enterprise SSO integration with Azure AD, Okta, etc.
 - Least-privilege access with permission sets
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